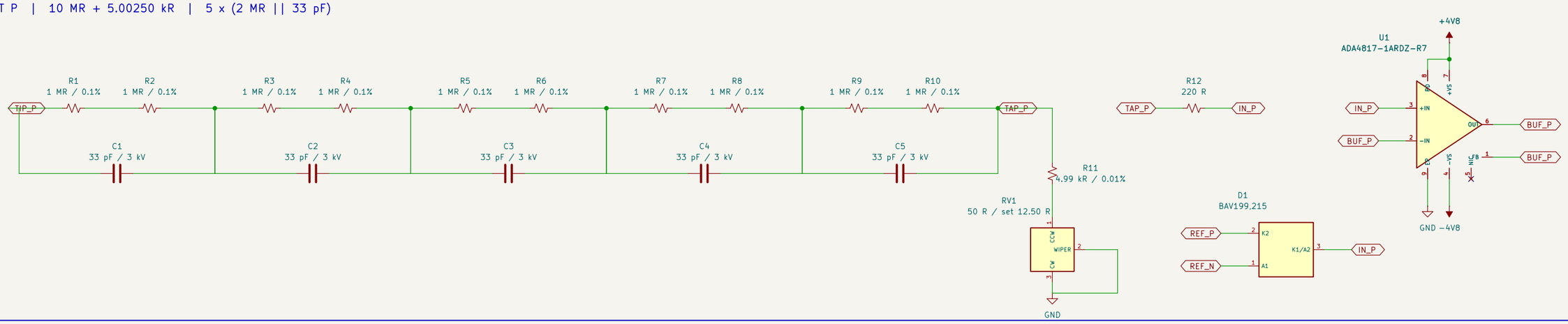
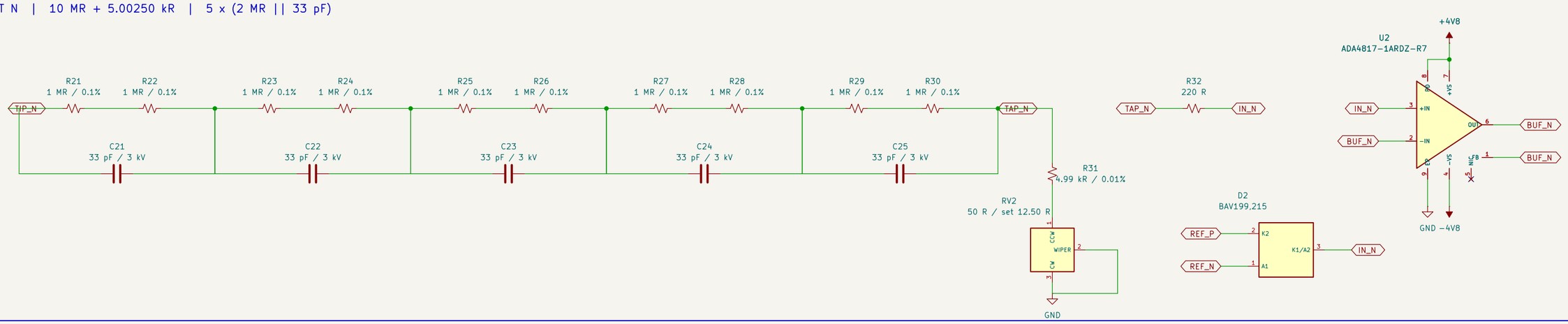
2000:1 INPUTS / matched geometry, fixed grounded shield

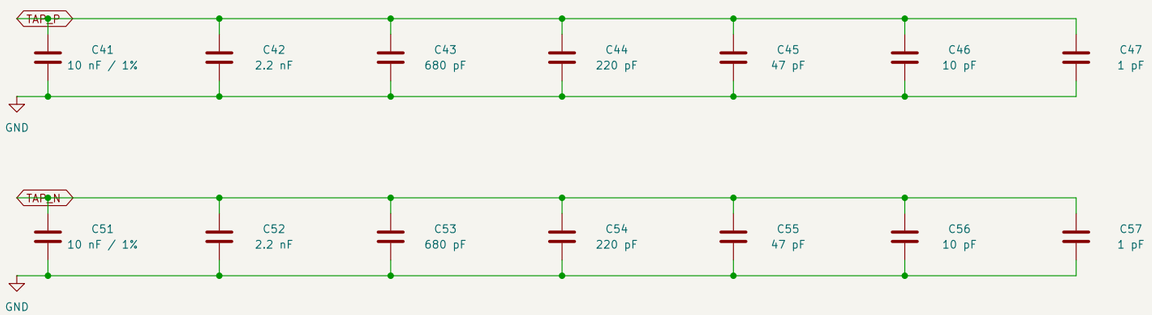
INPUT P | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)

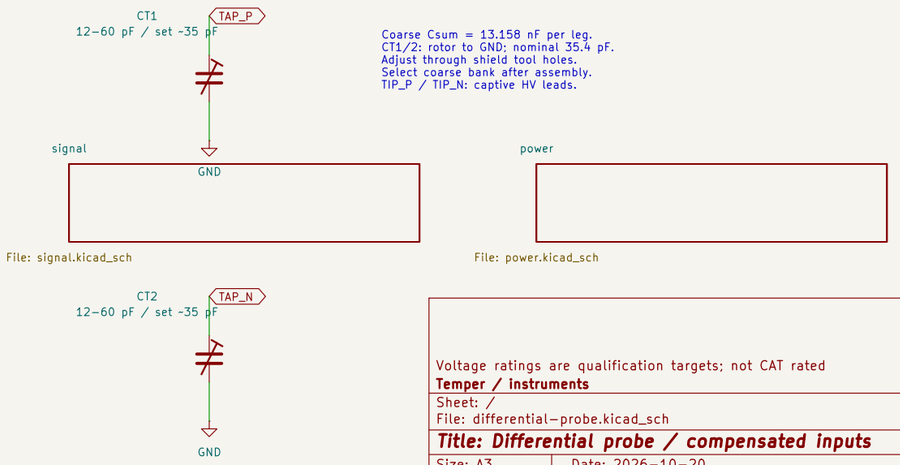


INPUT N | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)



AC COMPENSATION / coarse C0G bank + accessible fine trim





Coarse Csum = 13.158 nF per leg.
CT1/2: rotor to GND; nominal 35.4 pF.
Adjust through shield tool holes.
Select coarse bank after assembly.
TIP_P / TIP_N: captive HV leads.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /
File: differential–probe.kicad_sch

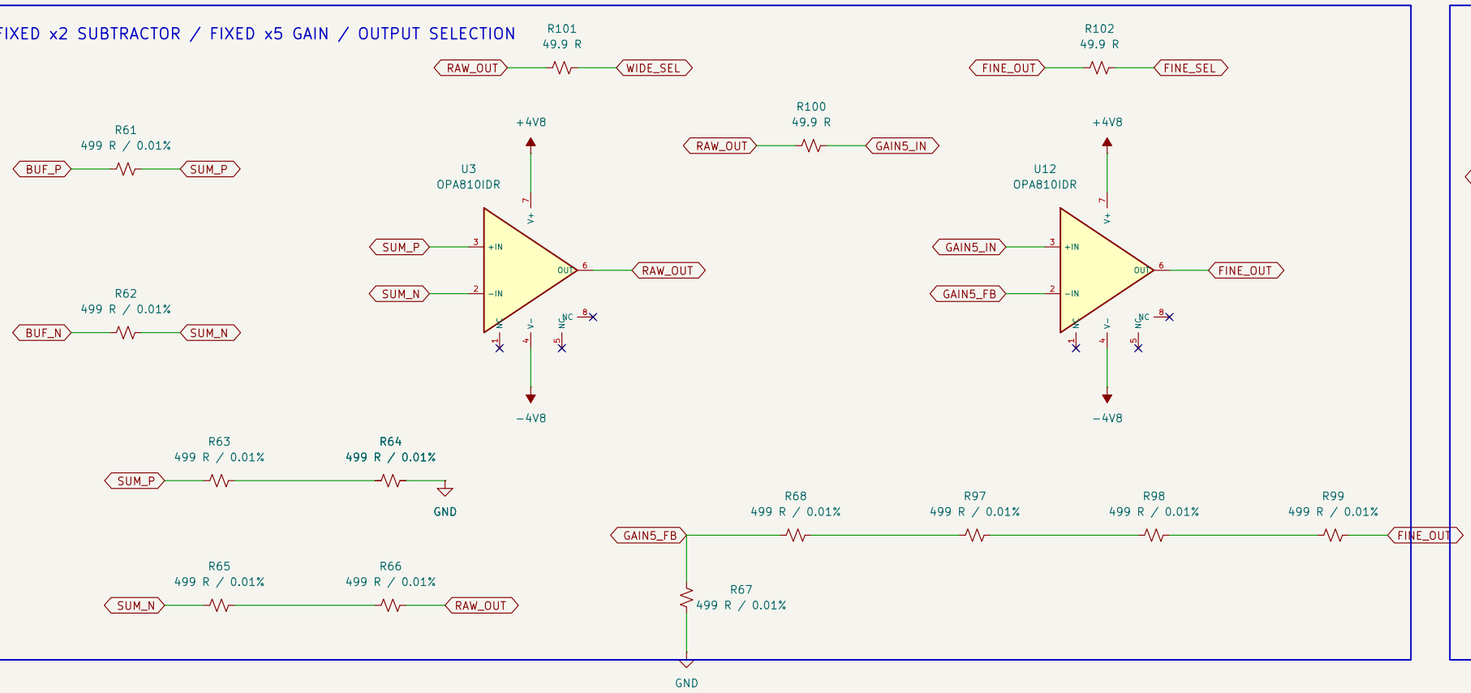
Title: Differential probe / compensated inputs

Size: A3	Date: 2026–10–20	**Rev: B prototype**
KiCad E.D.A. 10.0.4		Id: 1/3

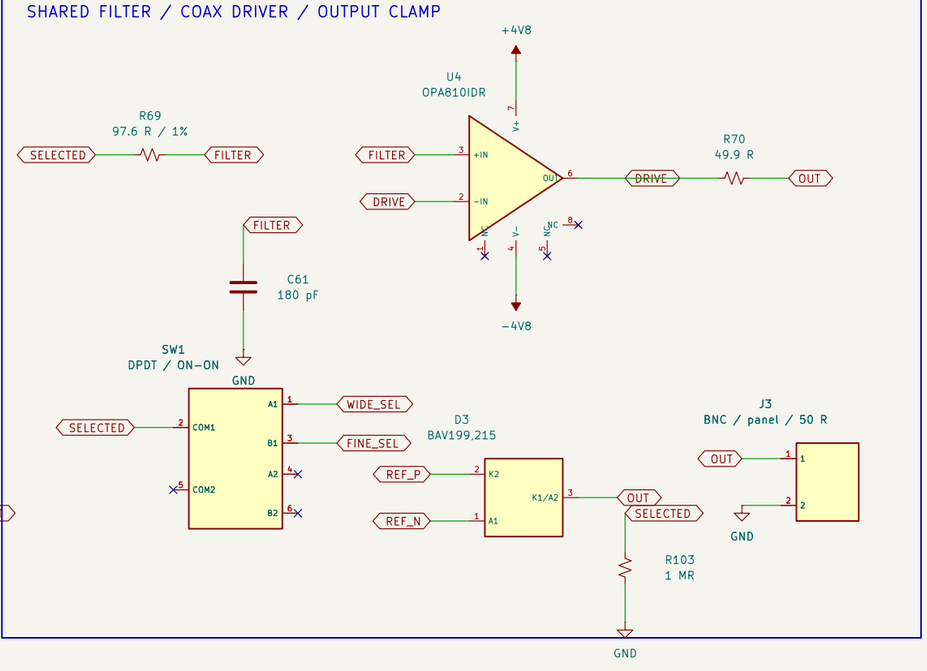
1000:1 / 200:1 OUTPUT / scope input = 1 MR

SW1 pin 2 selects 1 = 1000:1, 3 = 200:1. No switched feedback. Verify BBM contacts; allow settling during changes.

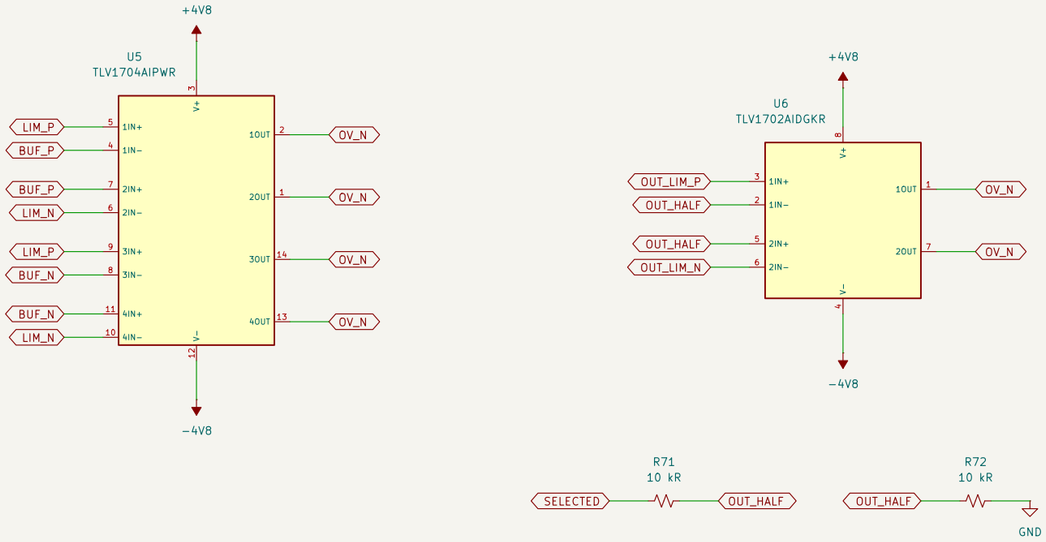
FIXED x2 SUBTRACTOR / FIXED x5 GAIN / OUTPUT SELECTION



SHARED FILTER / COAX DRIVER / OUTPUT CLAMP

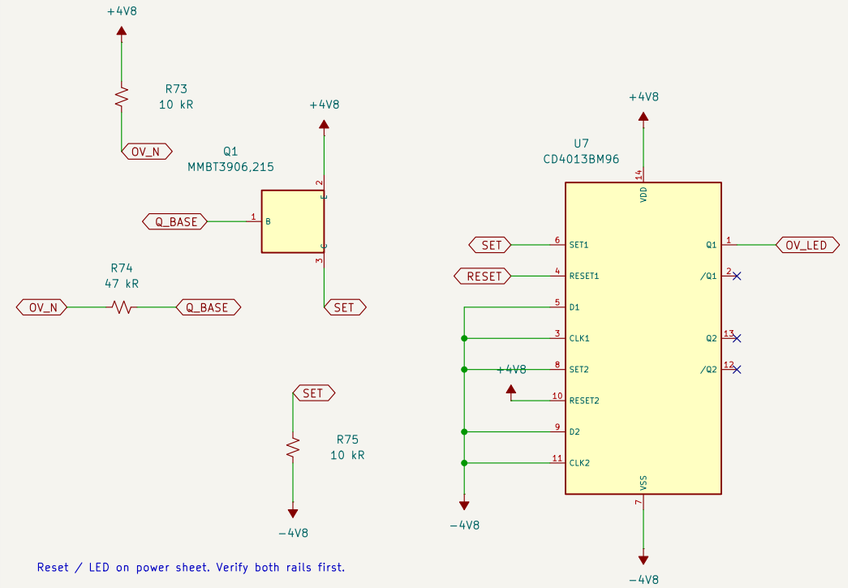


OPERATING-LIMIT WARNING / tips ~1406 V; selected signal ~1.266 V



TLV170x: 560 ns typical delay. Indicator is not a fast peak detector.
Unused x5 stage may saturate in 1000:1 range; selected signal alone drives output warning.

LATCHED OVERLOAD / RESET



Reset / LED on power sheet. Verify both rails first.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /signal/
File: signal.kicad_sch

Title: Differential probe / gain, output and overload

Size: A3	Date: 2026-10-20	**Rev: B prototype**
KiCad E.D.A. 10.0.4		Id: 2/3

BATTERY POWER / scope shield defines circuit ground

TWO 9 V BATTERIES / +/−4.8 V LOW-NOISE RAILS

BT1
9 V battery clip
BAT_P
GND

BT2
9 V battery clip
GND
BAT_N

SW2
POWER / DPDT / 5 A
BAT_P
BAT_N
GND
COM1
COM2
A1
B1
A2
B2
SW_P
SW_N

+BAT
D4
B140
SW_P

D5
B140
SW_N
−BAT

U8
ADP7142AUJZ-R7
+BAT
+BAT
VIN
EN
GND
VOUT
ADJ
+4V8
ADJ_0
GND

U9
ADP7182AUJZ-R7
−BAT
−BAT
VIN
EN
GND
VOUT
ADJ
−4V8
ADJ_1
GND

C71
10 uF / 25 V
+BAT
GND

C72
10 uF / 25 V
+4V8
GND

C73
10 uF / 25 V
−BAT
GND

C74
10 uF / 25 V
−4V8
GND

R81
30 kR
+4V8
ADJ_0

R82
10 kR
ADJ_0
GND

R83
29.4 kR
−4V8
ADJ_1

R84
10 kR
ADJ_1
GND

SHUNT CLAMPS / COMPARATOR THRESHOLDS

+4V8
R85
1 kR
REF_P
REF_P
U10
TL431BIDBZR
REF
K
A
GND

R87
25.5 kR
REF_P
LIM_P

R88
10 kR
LIM_P
GND

R89
29.4 kR
REF_P
OUT_LIM_P

R90
10 kR
OUT_LIM_P
GND

−4V8
R86
1 kR
REF_N
REF_N
GND
U11
TL431BIDBZR
REF
K
A
GND
REF_N

R91
25.5 kR
REF_N
LIM_N

R92
10 kR
LIM_N
GND

R93
29.4 kR
REF_N
OUT_LIM_N

R94
10 kR
OUT_LIM_N
GND

LOCAL DECOUPLING / RAIL REVERSAL DIODES

+4V8
U1
U2
U3
U4
U5
U6
U7
C81 100 nF / 50 V
C83 100 nF / 50 V
C85 100 nF / 50 V
C87 100 nF / 50 V
C89 100 nF / 50 V
C91 100 nF / 50 V
C93 100 nF / 50 V
C96 100 nF / 50 V
GND
C82 100 nF / 50 V
C84 100 nF / 50 V
C86 100 nF / 50 V
C88 100 nF / 50 V
C90 100 nF / 50 V
C92 100 nF / 50 V
C94 100 nF / 50 V
C97 100 nF / 50 V
−4V8

D6
B140
+4V8
GND

D7
B140
GND
−4V8

RESET / STATUS / POWER VALIDATION

SW3
RESET / NO
+4V8
RESET

R95
100 kR
RESET
−4V8

C95
100 nF / 50 V
+4V8
RESET

D8
1N4148W
−4V8
RESET

R96
10 kR
OV_LED
LED_A

D9
RED / OVERLOAD
LED_A
−4V8

Two 9 V alkaline batteries; center tap = GND.
Verify rails, reset, then connect DUT.
No charging / USB connection during measurement.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /power/
File: power.kicad_sch

Title: Differential probe / battery supply and references

Size: A3	Date: 2026-10-20	**Rev: B prototype**
KiCad E.D.A. 10.0.4		Id: 3/3